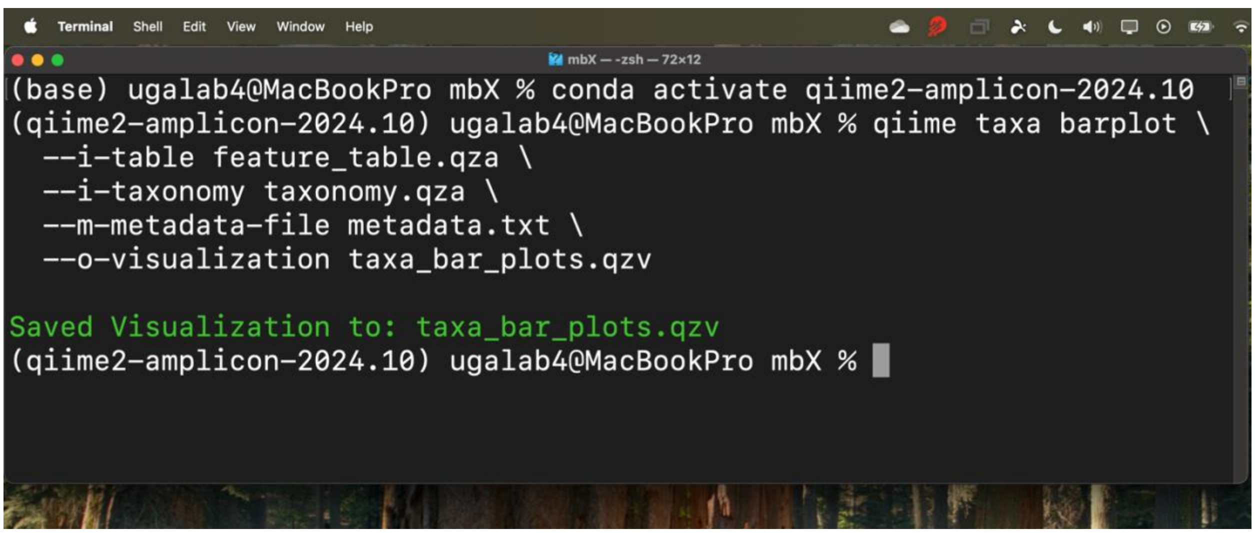
Easy step by step tutorial for microbiomeX

Step 1: Export the Taxa Barplot

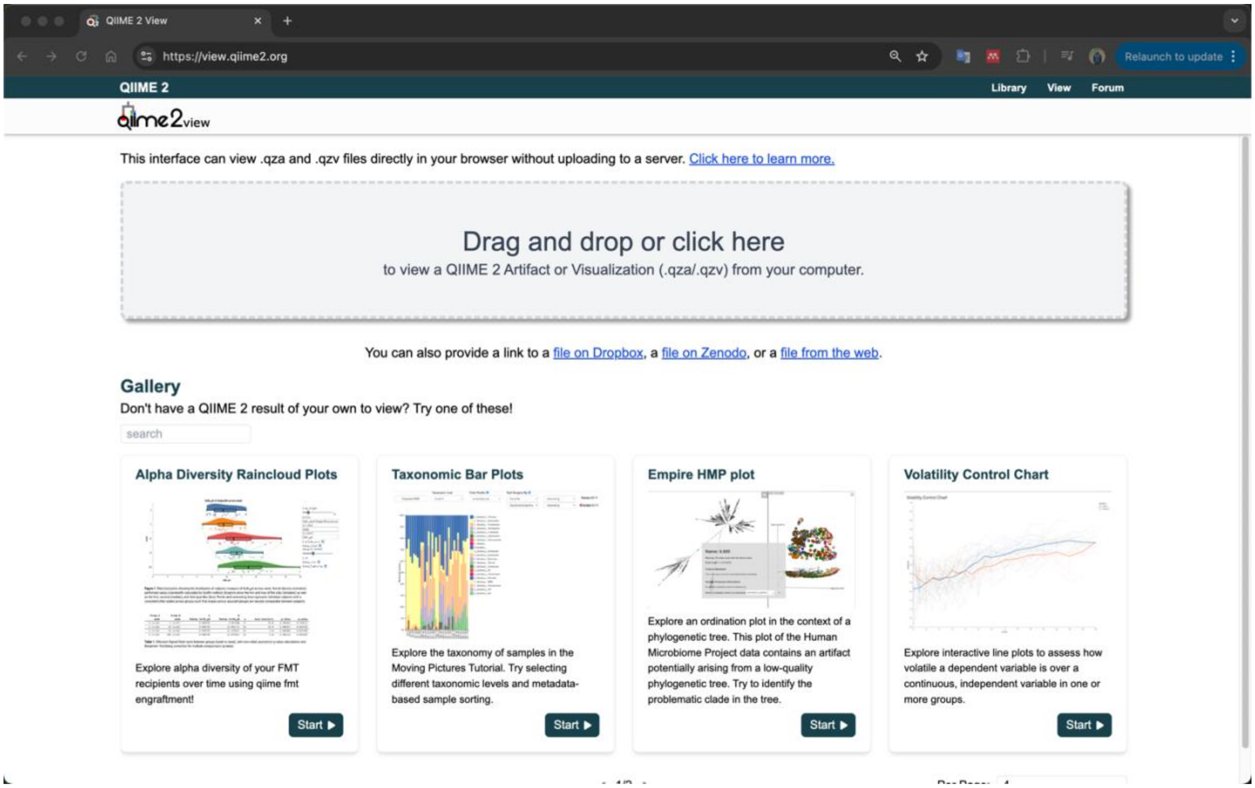
Run the following command in your terminal to export the taxa barplot from QIIME2:

```
qiime taxa barplot \
  --i-table feature_table.qza \
  --i-taxonomy taxonomy.qza \
  --m-metadata-file metadata.txt \
  --o-visualization taxa_bar_plots.qzv
```



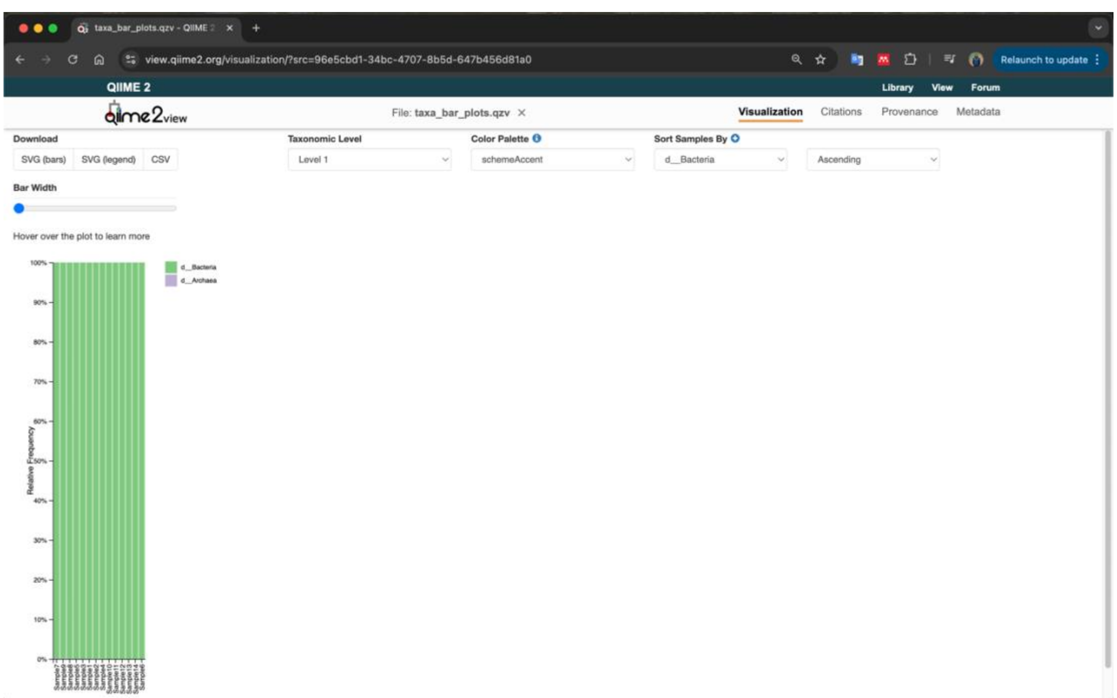
Step 2: Open the QIIME2 Viewer

Navigate to https://view.qiime2.org/.



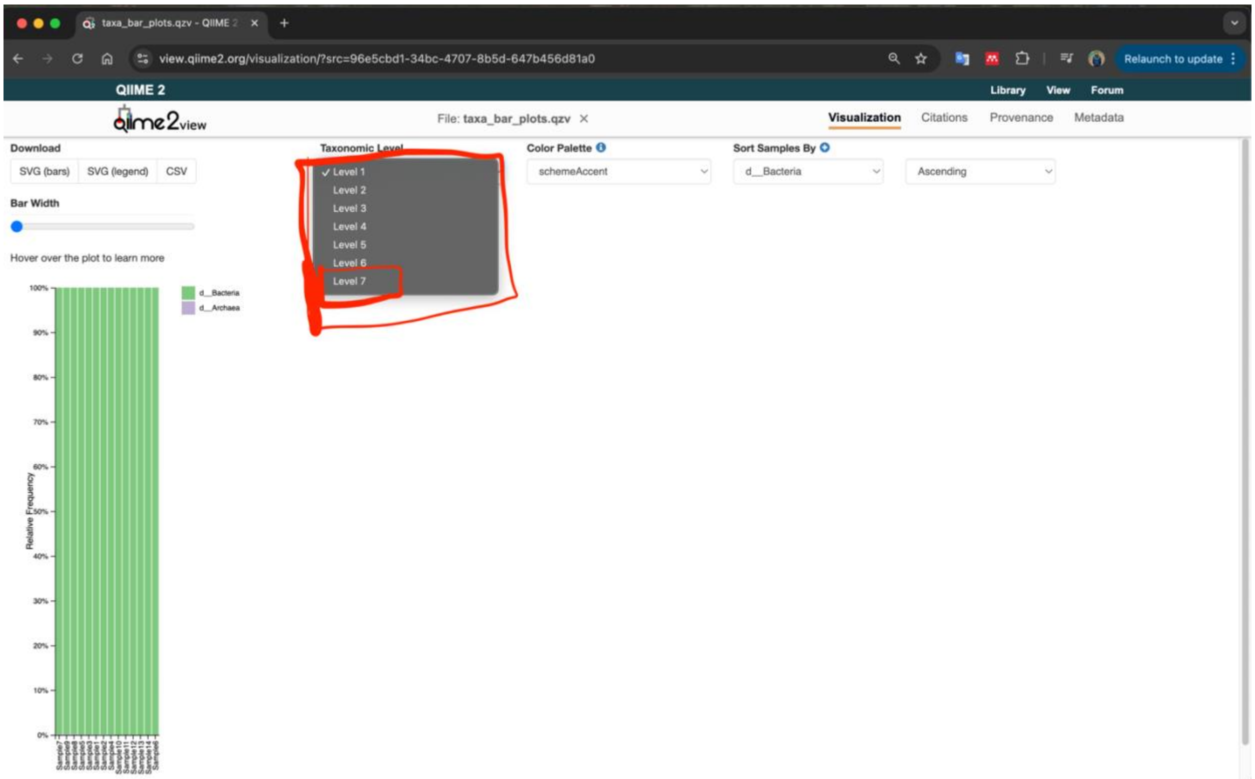
Step 3: Load Your Visualization File

Drag and drop the `taxa_bar_plots.qzv` file into the website. The visualization should appear immediately.



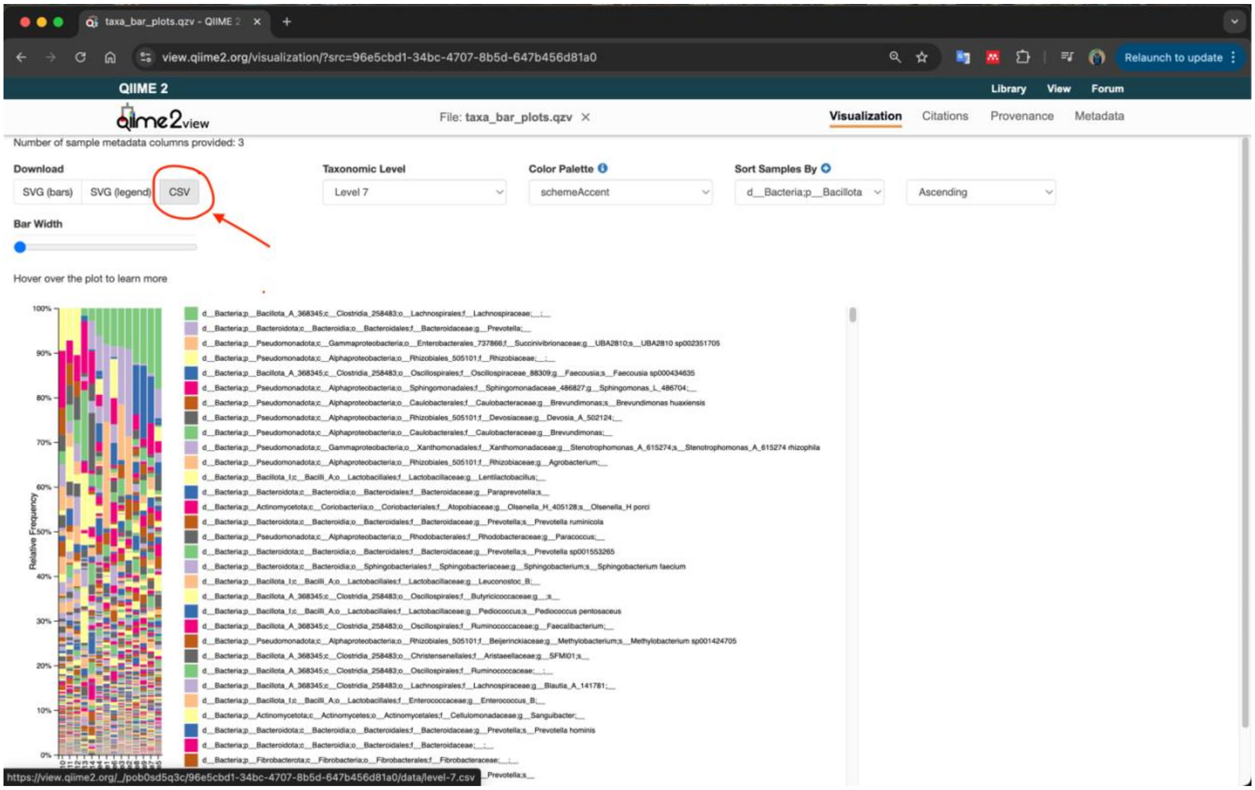
Step 4: Set the Taxonomic Level

Locate the dropdown menu labeled “Taxonomic Level” and select **Level 7**.

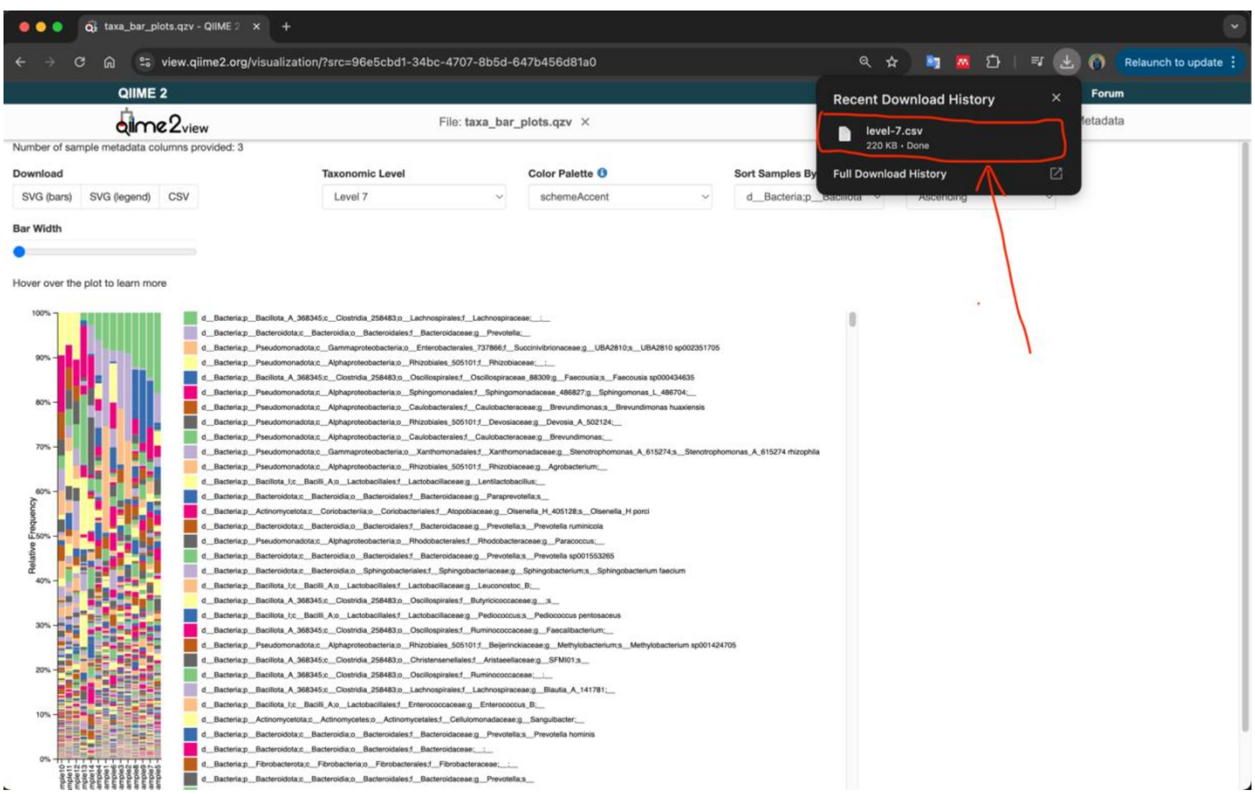


Step 5: Download the CSV File

Click on the **Download** option and choose **CSV**.



The file will be downloaded as `level-7.csv`.



Step 6: Organize Your Files

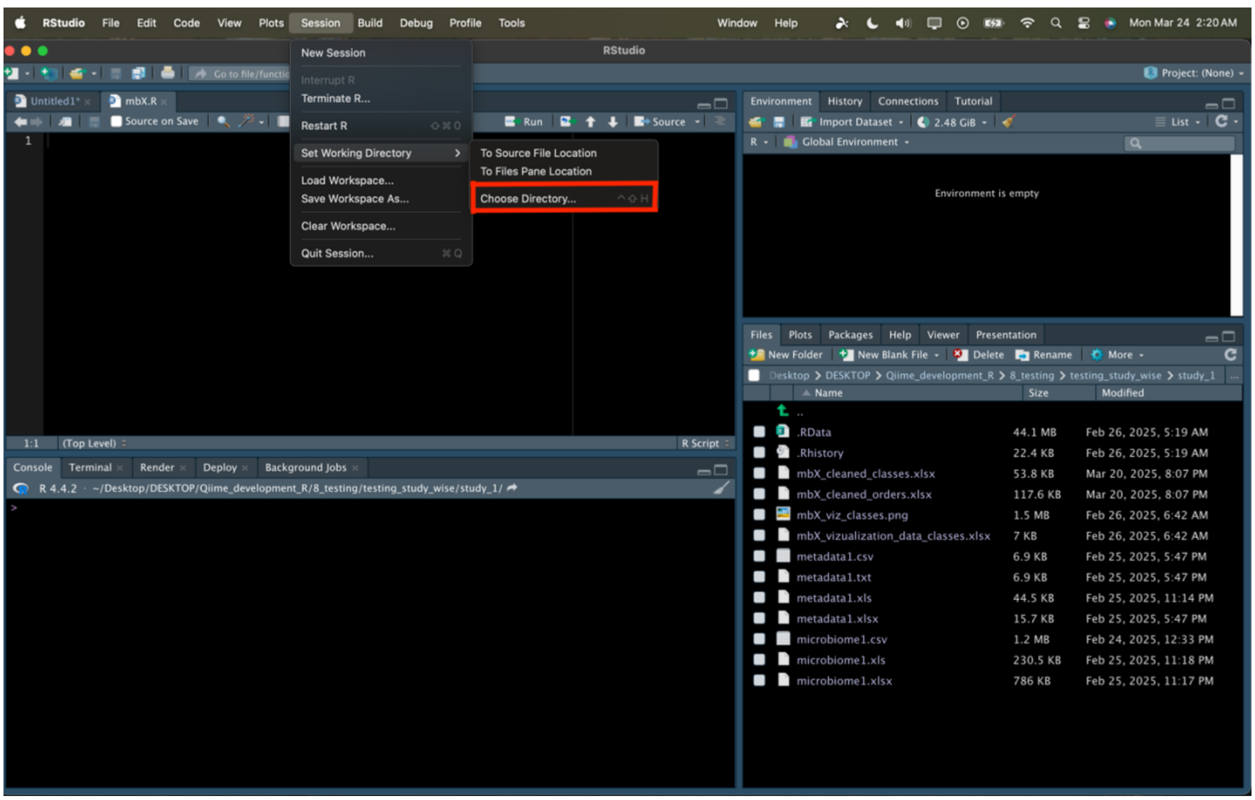
Place the downloaded `level-7.csv` file and the `metadata.txt` file (the same file used in QIIME2) into the same directory or folder of your choice.

Step 7: Open RStudio

Launch RStudio on your computer.

Step 8: Set Your Working Directory

Set the working directory in RStudio to the folder containing your `level-7.csv` and `metadata.txt` files.

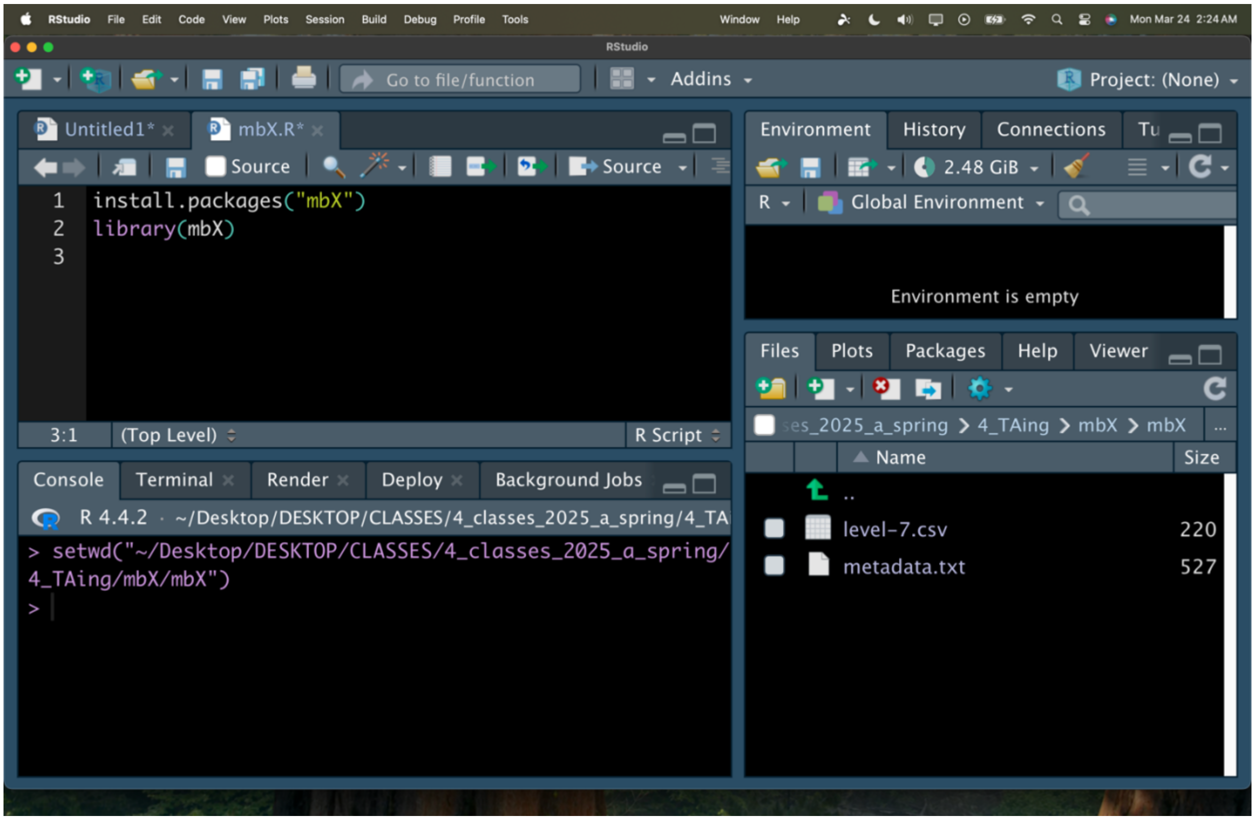


Step 9: Install and Load the mbX Package

Install the mbX package (if not already installed) and load it:

```
install.packages("mbX")

library(mbX)
```



Step 10: Run the ezclean Function

Use the `ezclean` function to process your data. In this example, the data is processed at the family level:

```
ezclean(
  microbiome_data = "level-7.csv",
  metadata = "metadata.txt",
  level = "f"  # Use "f" for family; change to "g" for genus, "s" for species, etc.
)
```

Or using the simplified version:

```
ezclean("level-7.csv", "metadata.txt", "f")
```

After running `ezclean`, the cleaned output (e.g., `mbX_cleaned_families.xlsx`) will be saved in your working directory.

Step 11: Run the ezviz Function

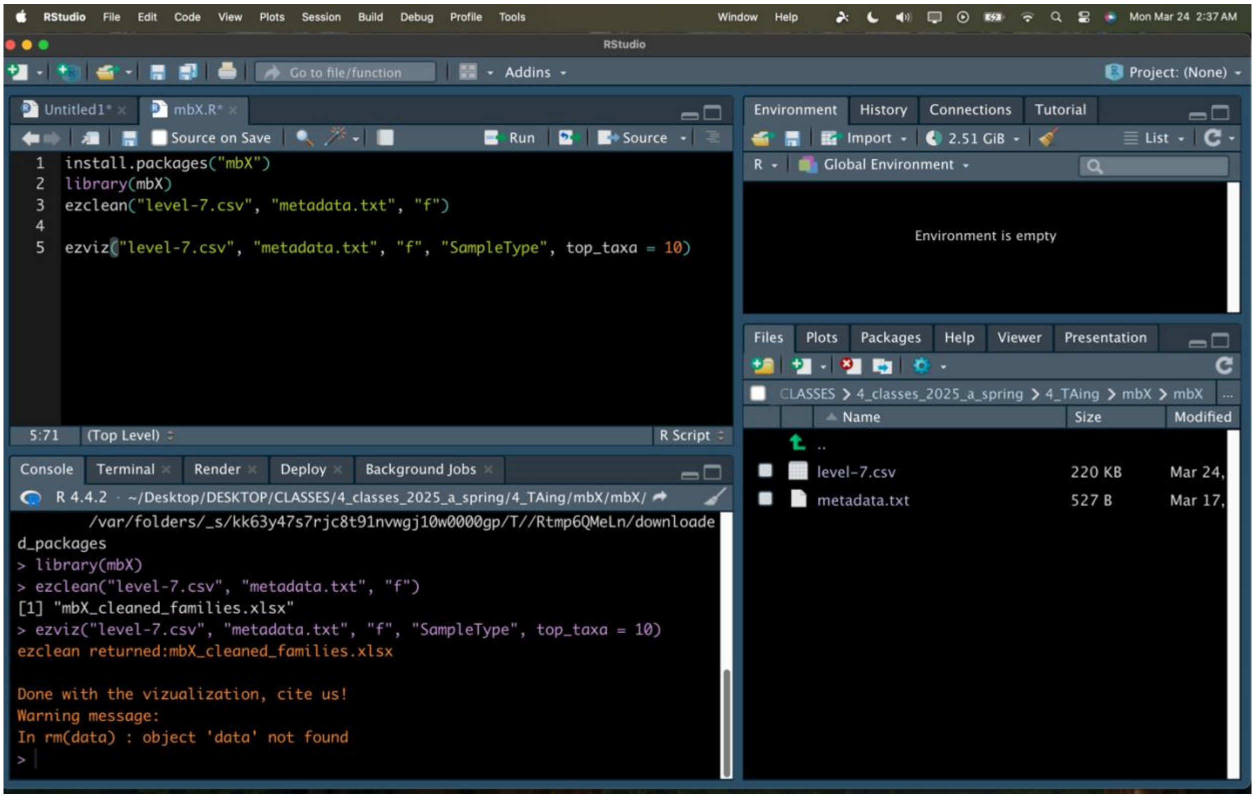
Generate publication-ready visualizations using the `ezviz` function:

```
ezviz(
  microbiome_data = "level-7.csv",
  metadata = "metadata.txt",
  level = "family",        # Or your chosen taxonomic level (e.g., "genus")
  selected_metadata = "SampleType",  # Replace with the column name used for grouping
  top_taxa = 10           # Alternatively, use 'threshold' if preferred
)
```

Or using the simplified version:

```
ezviz("level-7.csv", "metadata.txt", "f", "SampleType", top_taxa = 10)
```

After running `ezviz`, the processed visualization data (Excel file) and a high-resolution PDF plot (e.g., `mbX_viz_families.pdf`) will be saved in your working directory.


RStudio File Edit Code View Plots Session Build Debug Profile Tools Window Help
Mon Mar 24 2:37 AM
RStudio
Go to file/function
Addins
Project: (None)
Untitled1*
mbX.R*
Source on Save
Run
Source
install.packages("mbX")
library(mbX)
ezclean("level-7.csv", "metadata.txt", "f")
ezviz("level-7.csv", "metadata.txt", "f", "SampleType", top_taxa = 10)
5:71
(Top Level)
R Script
Console
Terminal
Render
Deploy
Background Jobs
R 4.4.2 · ~/Desktop/DESKTOP/CLASSES/4_classes_2025_a_spring/4_TAing/mbX/mbX/
/var/folders/_s/kk63y47s7rjc8t91nvwgj10w0000gp/T//Rtmp6QMeLn/downloaded_packages
> library(mbX)
> ezclean("level-7.csv", "metadata.txt", "f")
[1] "mbX_cleaned_families.xlsx"
> ezviz("level-7.csv", "metadata.txt", "f", "SampleType", top_taxa = 10)
ezclean returned:mbX_cleaned_families.xlsx
Done with the vizualization, cite us!
Warning message:
In rm(data) : object 'data' not found
>
Environment
History
Connections
Tutorial
Import
2.51 GiB
List
R
Global Environment
Environment is empty
Files
Plots
Packages
Help
Viewer
Presentation
CLASSES > 4_classes_2025_a_spring > 4_TAing > mbX > mbX
Name
Size
Modified
..
level-7.csv
220 KB
Mar 24,
metadata.txt
527 B
Mar 17,